

Week 4 Recitation

TA Suha Kim

ECON8711 (Micro 1A)

Sep 13, 2024

Production Functions

Determining concavity of a function

CES Production: $f(x_1, x_2) = (a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}}$
(where $\rho < 1, a_1 > 0, a_2 > 0$)

Is this a concave function?

$$\begin{aligned} & \max_{x \in \mathbb{R}_+} pf(x) - wx \\ & \quad \hookrightarrow \text{2nd deriv.} \\ & \quad \quad \underbrace{pf''(x)}_{\substack{\uparrow \\ \text{concave opt. problem}}} \leq 0 \end{aligned}$$

Determining Concavity

"Cookbook" procedure

e.g.

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$A_1^\pi = a$
 $A_3^\pi = A^\pi$
 $A_2^\pi = \begin{pmatrix} a & b \\ d & e \end{pmatrix}$ ✓

Denote by

- ✓ • $\pi = (\pi_1, \pi_2, \dots, \pi_n)$: permutation of the integers $\{1, \dots, n\}$
- A^π : the symmetric $n \times n$ matrix applying the permutation π to rows and columns of A
- for $k \in \{1, \dots, n\}$, A_k^π : $k \times k$ submatrix of A^π obtained by retaining only the first k rows and columns

Semi-definiteness

A **symmetric** $n \times n$ matrix A is

- 1 positive semidefinite iff $|A_k^\pi| \geq 0$ for all $k \in \{1, 2, \dots, n\}$ and for all $\pi \in \Pi$.
- 2 negative semidefinite iff $\underbrace{(-1)^k}_{\text{wavy line}} |A_k^\pi| \geq 0$ for all $k \in \{1, 2, \dots, n\}$, and for all $\pi \in \Pi$

✓

$$-|a| \geq 0, \quad | \begin{pmatrix} a & b \\ d & e \end{pmatrix} | \geq 0, \quad -| \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} | \geq 0$$

Determining Concavity

“Cookbook” procedure

Concavity(Convexity) of a C^2 function

Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a C^2 function, where $\mathcal{D} \subseteq \mathbb{R}^n$ is open and convex. Then,

- ① f is concave on $\mathcal{D}$ iff $D^2 f(x)$ is a negative semidefinite matrix for all $x \in \mathcal{D}$.
- ② f is convex on $\mathcal{D}$ iff $D^2 f(x)$ is a positive semidefinite matrix for all $x \in \mathcal{D}$.
- ③ If $D^2 f(x)$ is a negative definite matrix for all $x \in \mathcal{D}$, then f is strictly concave on $\mathcal{D}$.
- ④ If $D^2 f(x)$ is a positive definite matrix for all $x \in \mathcal{D}$, then f is strictly convex on $\mathcal{D}$.

Determining Concavity

“Cookbook” procedure

$$\checkmark f(x_1, x_2) = \ln(x_1 + x_2) \\ \neq \ln(2x_1 + x_2)$$

CES Production: $f(x_1, x_2) = (a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}}$
(where $\rho < 1, a_1 > 0, a_2 > 0$)

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix}$$

1 ✓ Hessian of the function f ?

2 Negative semidefiniteness of the minors of Hessian?

Determining Concavity

Another way

$$\underline{\rho < 1} \quad \rho < 0$$

CES Production: $f(x_1, x_2) = (a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}} = h(g(x_1, x_2))$

where $h(x) = x^{\frac{1}{\rho}}$, $g(x_1, x_2) = a_1 x_1^\rho + a_2 x_2^\rho$

$$H = \begin{pmatrix} 0 & \rho a_1 x_1^{\rho-1} & \rho a_2 x_2^{\rho-1} \\ \rho a_1 x_1^{\rho-1} & \rho(\rho-1)a_1 x_1^{\rho-2} & 0 \\ \rho a_2 x_2^{\rho-1} & 0 & \rho(\rho-1)a_2 x_2^{\rho-2} \end{pmatrix}$$

- ① g is quasi-concave

$$\checkmark \frac{\partial g}{\partial x_1} = \rho a_1 x_1^{\rho-1}$$

$$\frac{\partial g}{\partial x_2} = \rho a_2 x_2^{\rho-1}$$

$$-|C_1| = \rho^2 a_1^2 x_1^{2(\rho-1)} > 0$$

- ★ ② Monotone transformation of a (quasi)-concave function must also be quasi-concave

$$\begin{aligned} |C_2| &= -\rho^3(\rho-1)a_1 a_2^2 x_1^{\rho-2} x_2^{2\rho} \\ &\quad - \rho^3(\rho-1)a_1^2 a_2 x_1^{2(\rho-1)} x_2^{\rho-2} \\ &> 0 \end{aligned}$$

- ★ ③ A quasi-concave function that is HD1 is concave. ↙

$$\begin{array}{ccc} \text{concave} & \Rightarrow & \text{quasi-concave} \\ & \nwarrow & \\ & \text{HD} & \end{array}$$

Determining Quasi-Concavity

“Cookbook” procedure

For $k = 1, 2, \dots, n$, let $C_k(x)$ be the $(k+1) \times (k+1)$ matrix defined as $(C_n(x): \text{Bordered Hessian})$

$$C_k(x) = \begin{bmatrix} 0 & \frac{\partial f}{\partial x_1}(x) & \cdots & \frac{\partial f}{\partial x_k}(x) \\ \frac{\partial f}{\partial x_1}(x) & \frac{\partial^2 f}{\partial x_1^2}(x) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_k}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_k}(x) & \frac{\partial^2 f}{\partial x_k \partial x_1}(x) & \cdots & \frac{\partial^2 f}{\partial x_k^2}(x) \end{bmatrix}$$

Quasi-Concavity(Quasi-Convexity) of a C^2 function

Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a C^2 function, where $\mathcal{D} \subseteq \mathbb{R}^n$ is open and convex. Then,

- ① If f is quasi-concave on $\mathcal{D}$, then we have $(-1)^k |C_k(x)| \geq 0$ for $k \in \{1, 2, \dots, n\}$.
- ② If $(-1)^k |C_k(x)| > 0$ for all $k \in \{1, 2, \dots, n\}$, then f is quasi-concave on $\mathcal{D}$.

Determining Concavity

Another way

$$H = \left(\begin{array}{c} \text{---} \end{array} \right)$$

Another example: $f(y_1, y_2, y_3) = \sum_{k=1}^3 (u_k y_k - \lambda y_k \ln y_k)$,
where $\lambda > 0$ and $u_k \in \mathbb{R}$

- ① $\underbrace{g_k(y_k)} = u_k y_k - \lambda y_k \ln y_k$ is concave ($k = 1, 2, 3$).

$$\frac{\partial g_k}{\partial y_k} = u_k - \lambda(\ln y_k + 1) \quad , \quad \frac{\partial^2 g_k}{\partial y_k^2} = -\frac{\lambda}{y_k} < 0$$

- ✓ ② The sum of concave functions is also concave.
 $f(y_1, y_2, y_3) = g_1(y_1) + g_2(y_2) + g_3(y_3)$

Profit function

Properties

Properties of profit function $\pi(p) = \sup_{z \in Z} \{p \cdot z \mid z \in Z\}$

- ① $\pi(p)$ is increasing in output prices, and decreasing in input prices
- ② $\pi(p)$ is HD1: $\pi(tp) = t\pi(p)$ for $t \geq 0$
- ③ $\pi(p)$ is convex in p
- ④ $\pi(p)$ is continuous in p

Prove #2: $\pi(p)$ is HD1

Take any $p > 0, t > 0$

i) $(\geq)$ ✓

Let $\varepsilon > 0$ be given

$\exists \tilde{z} \in Z$ s.t. $p \cdot \tilde{z} \geq \pi(p) - \varepsilon$

$\pi(tp) \geq (tp) \cdot \tilde{z} = t p \cdot \tilde{z} \geq t(\pi(p) - \varepsilon) = \frac{1}{t} \pi(tp)$

$\therefore \pi(tp) \geq t\pi(p)$ ✓

ii) $(\leq)$ ✓

$p' = tp, t' = \frac{1}{t}$

$t p' = p$

using (i) $\pi(p) \geq \frac{1}{t'} \pi(\frac{p'}{t'})$

$\pi(p) \geq \frac{1}{\frac{1}{t}} \pi(\frac{tp}{\frac{1}{t}})$

$= t \pi(p) - t \varepsilon / t = t \pi(p) - \varepsilon$

$$\pi(p') \geq \frac{1}{t} \pi\left(\frac{p'}{t}\right)$$

$$= \frac{1}{t'} \pi(t' p')$$

$$\underline{t' \pi(p')} \geq \underline{\pi(t' p')}$$

(Revisited Note!)

Prove #2: $\pi(p)$ is HD1

$$\text{Show } \sup_{z \in \mathbb{Z}} \{t p \cdot z \mid z \in \mathbb{Z}\} = t \sup_{z \in \mathbb{Z}} \{p \cdot z \mid z \in \mathbb{Z}\}.$$

$\pi(tp)$ $t\pi(p)$

pf) Take any $p > 0$, $t > 0$.
 $\in \mathbb{R}^k$ $\in \mathbb{R}$

(i) ($\geq$)

Let $\frac{\epsilon}{t} > 0$ be given.

Then $\exists \tilde{z} \in \mathbb{Z}$ s.t. $p \cdot \tilde{z} \geq \pi(p) - \frac{\epsilon}{t}$

$$\begin{aligned} \pi(tp) &\geq t p \cdot \tilde{z} \quad (\pi(tp) \text{ is the supremum}) \\ &\geq t(\pi(p) - \frac{\epsilon}{t}) \\ &= t\pi(p) - t\frac{\epsilon}{t}. \end{aligned}$$

$$\therefore \pi(tp) \geq t\pi(p).$$

(ii) ($\leq$)

Let $p' \equiv \frac{p}{t}$, $t' \equiv \frac{1}{t}$. (Since p, t are arbitrary positive, $p' > 0$, $t' > 0$ are also arbitrary.)
 $t'p' = p$.

Using the result above in (i), $\pi(tp) \geq t\pi(p)$,

$$\begin{aligned} \pi(p') &\geq \frac{1}{t'} \pi\left(\frac{p'}{t'}\right) \\ &= \frac{1}{t'} \pi(t'p'). \end{aligned}$$

$$\therefore t'\pi(p') \geq \pi(t'p').$$

Hence, from (i) & (ii), for $\forall p, t > 0$, $\pi(tp) = t\pi(p)$.